

Simulation Homework

Airline Revenue Management.

Alpha Airlines (AA) has ordered a new fleet of DC-717s. At this stage of the contract, Alpha's operations manager must specify the seating configuration on the aircraft that will be used on the Boston-Atlanta-Chicago- Boston circuit. Alpha flies this route once each day.

The configuration decision involves specifying how many rows will be allocated for first class and how many for tourist class. If the aircraft were configured entirely of tourist rows (containing six seats each), there would be forty rows. First-class seats are wider and afford more legroom, so that, in order to make room for one first-class row (containing four seats), two tourist rows must be removed. Thus, conversion from tourist to first-class seating involves the loss of some seats, but the conversion may be appealing because the revenues are higher for first-class passengers than for tourist passengers (see Table 1).

A perfect match between the configuration and the demand for seats is seldom possible. Historical data suggest a probability distribution of demand for seats on each leg (as detailed in Table 2). There is another distribution for the fraction of demand that corresponds to first-class seats (Table 3), which seems to apply on all legs, although the fraction that occurs in one market on any day is independent of the fraction in the other markets. Finally, there is some chance that all seats in either seating category will be booked on a given leg when demand for that category occurs. Under present management policies, such demand is simply lost to competitors. In particular, excess first class demand is not directed to fill empty tourist seats and excess tourist demand is not directed to fill empty first class seats.

The fixed cost of operating the full circuit is \$100,000 per day. Alpha Airlines is seeking a profit-maximizing configuration.

Table 1. Revenue per seat

	First Class	Tourist
Bos-Atl	\$400	\$175
Atl-Chi	\$400	\$150
Chi-Bos	\$450	\$200

Table 2. Distribution of total demand for seats

	Minimum	Most likely	Maximum
Bos-Atl	160	180	220
Atl-Chi	140	200	240
Chi-Bos	150	200	225

Table 3. Distribution of fraction first class

Fraction	5%	12%	15%
Probability	0.2	0.5	0.3

Problem 1.

Suppose that you allocated four rows for the first-class passengers. How many seats would be available in the first class for this configuration?

How many seats would be available in the tourist class for this configuration?

Problem 2.

Continue operating with the four first class rows configuration.

Suppose that on a particular day, BOS-ATL demand is 200 and fraction of first-class demand is 12%. ATL-CHI demand is 175 and fraction of first-class demand is 5%. CHI-BOS demand is 200 and fraction of first-class demand is 15%.

What is the revenue the airline collected on BOS-ATL-CHI-BOS circuit on that day?

What is the profit of the airline on the same circuit that day?

Assume that fractional demand is possible.

As your responses, please you enter numbers with two decimal points precision and without commas. For example, instead of 123,456.789 you should enter 123456.79

Problem 3.

Suppose you allocated three rows for first class passengers.

For this configuration, what is the airline's expected profit per day?

Assume that fractional demand is possible. Please run 10,000 simulation trials. Please enter a number with the precision of zero decimal points. That is instead of 123,456.789 please enter 123457.

Problem 4.

For the configuration with three rows allocated to the first-class passengers, what proportion of the days will airline at least break even?

Please enter your answer as a percentage (but without percentage sign) with the prevision of 2 decimal points. That is, instead of 0.12345 or 12.345% please enter 12.35

Problem 5.

What is the airline's optimal expected profit?

Please enter your answer as a number with the precision of zero decimal points. That is, instead of 123,456.789 please enter 123457

Problem 6.

To achieve the optimal expected profit, what is the number of rows that must be allocated to the first-class passengers?

Please enter your answer as a number (an integer).